

Learning to Predict Without Knowing: Implicit Priors Under Ambiguous Sensory History

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Pod:unkulunkulu/Bias in Progress

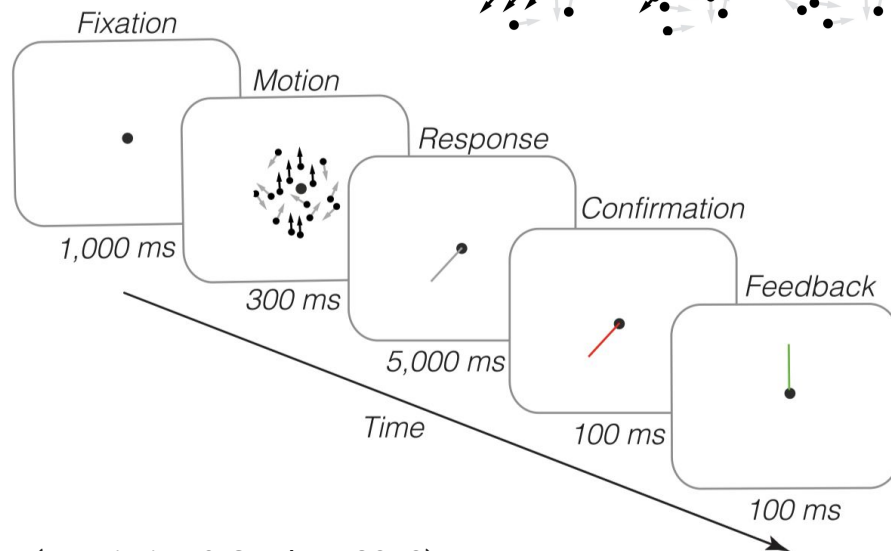
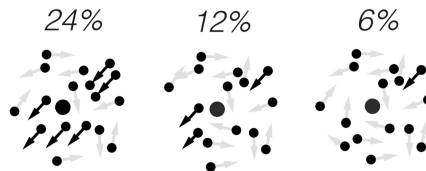
Blur Can't Hide McDonald's



When experience is unclear, how expectations are formed?

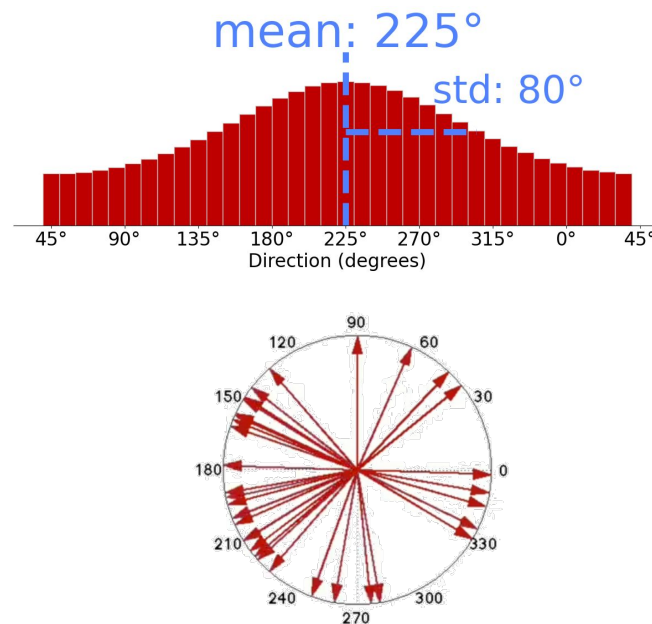
Task Design

Motion Coherence



(Laquittaine & Gardner, 2018)

Motion Direction Distribution (prior)



Hypothesis & Approach

Task:

Each run unit consists of **same 226 trials, of which:**

1) Motion Coherence:

(6%=109) + (12%=75) + (24%=42)

2) Motion Direction:

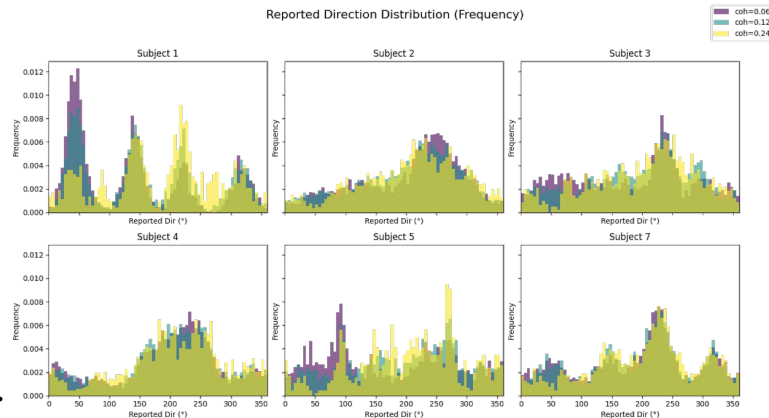
- start, end, step = [5,355,10];
- mean = 225° , std = 80°

Hypothesis:

Participants implicitly learn the frequent 225° direction.

Subjects:

- Six subjects completed varying numbers of runs (24 to 66).
- Observed highly diverse response strategies, while **most subjects show a concentration around 225° .**



Two approaches to test:

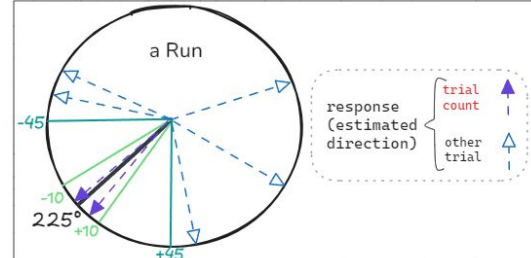
•Bottom-up : Random Forest classifier

Data-driven: “Can we directly detect the learning by analyzing raw behavior data?”

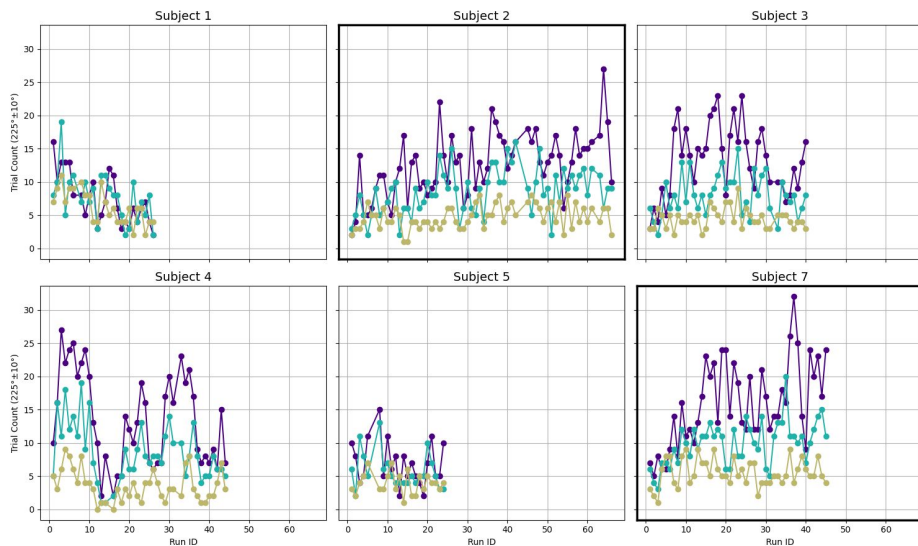
•Top-down: Online Bayesian model

Model-driven: “How do they update their prior beliefs over time based on trial feedback?”

Behavioral Data Statistics

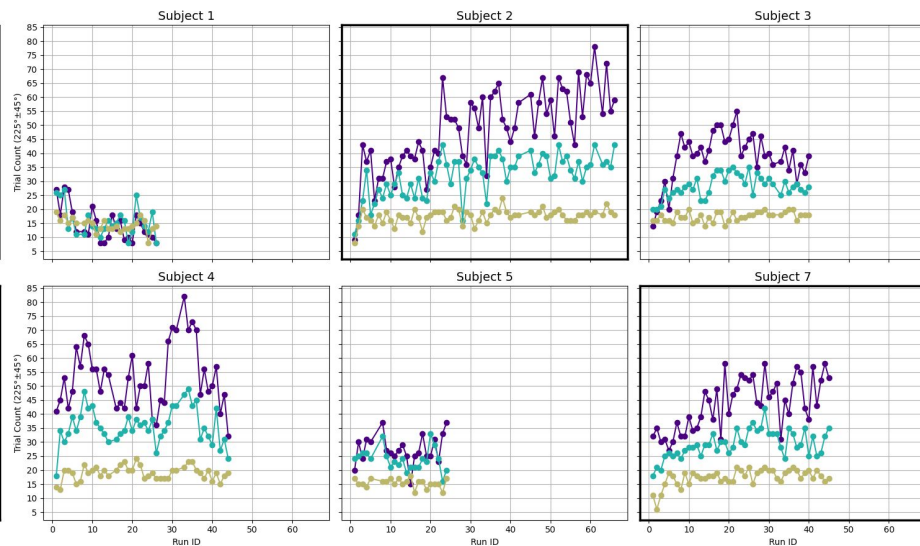


Count of Estimated Direction in Prior Range ($225^\circ \pm 10^\circ$) across Runs by Subject (filtered for runs with full 226 trials)



Prior Range: 225 ± 10 degree

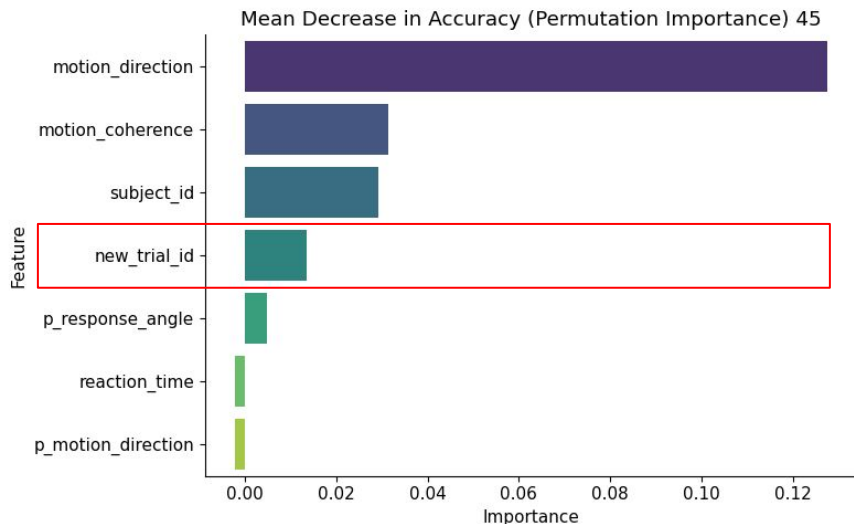
Count of Estimated Direction in Prior Range ($225^\circ \pm 45^\circ$) across Runs by Subject (filtered for runs with full 226 trials)



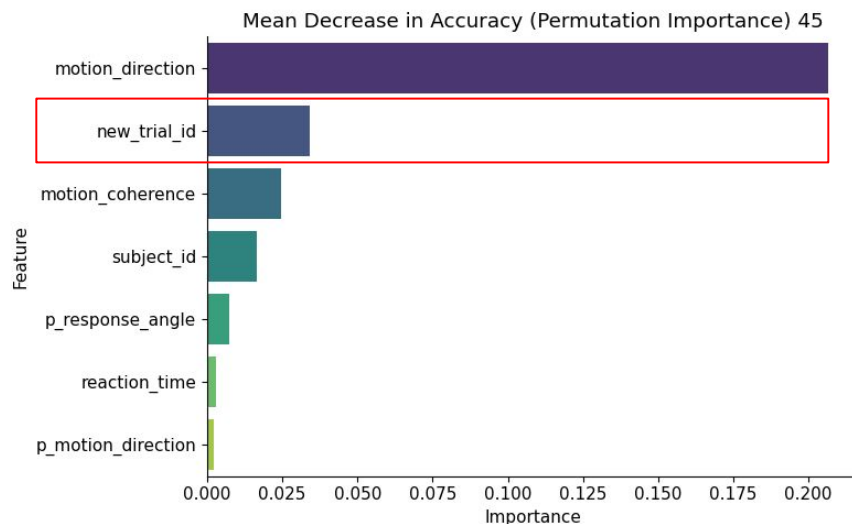
Prior Range: 225 ± 45 degree

Random Forest Analysis

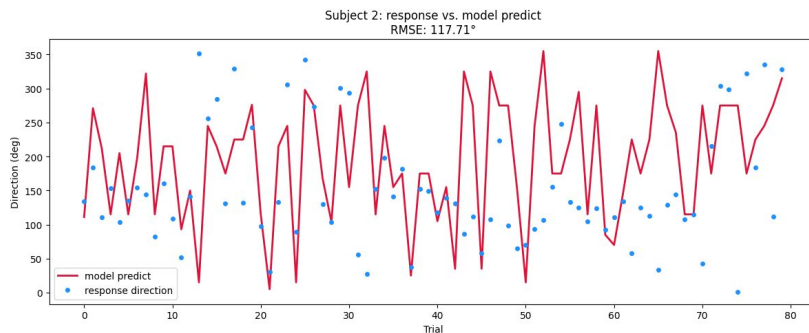
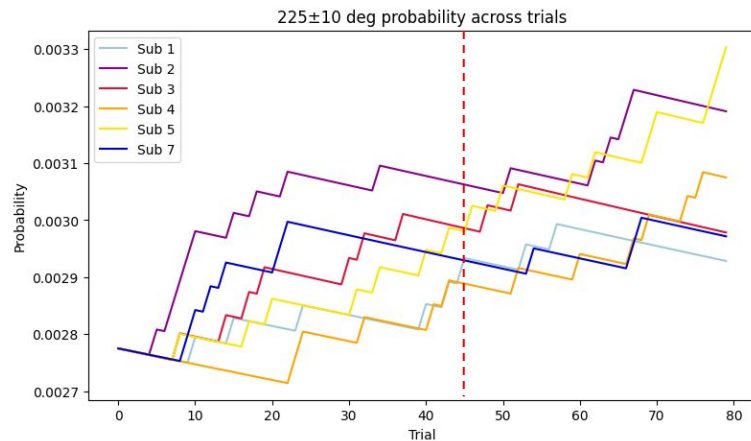
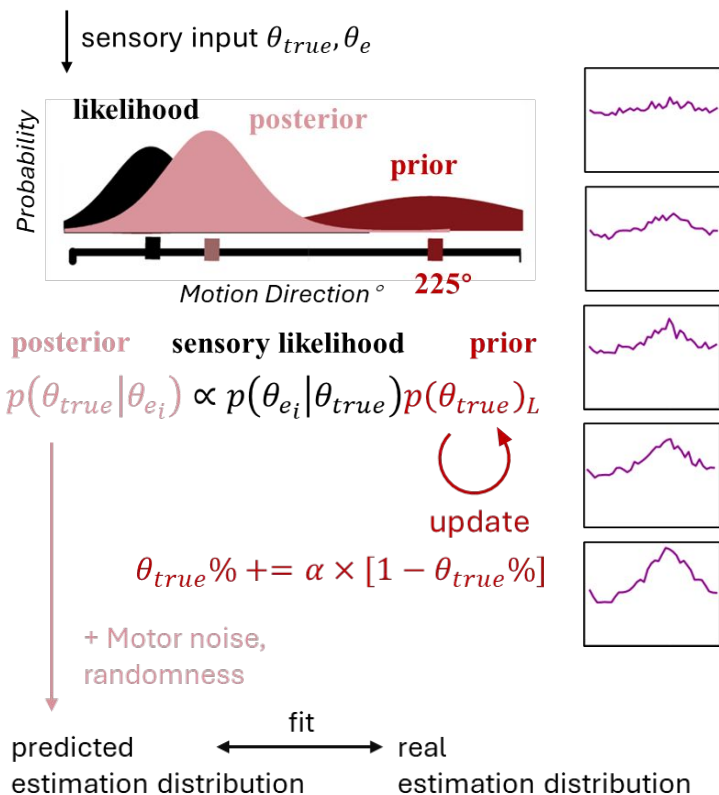
First 10 RUN



Last 10 RUN



Online Bayesian Model



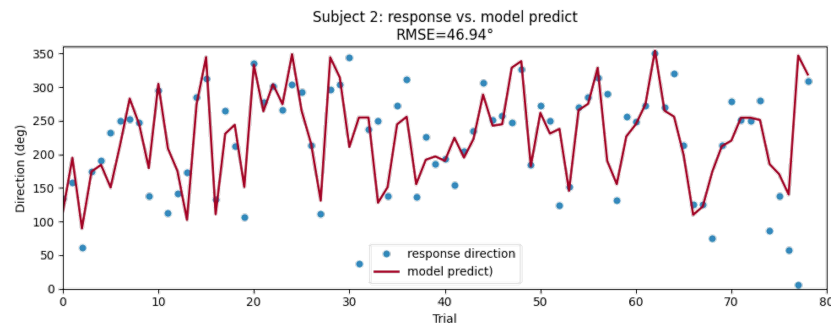
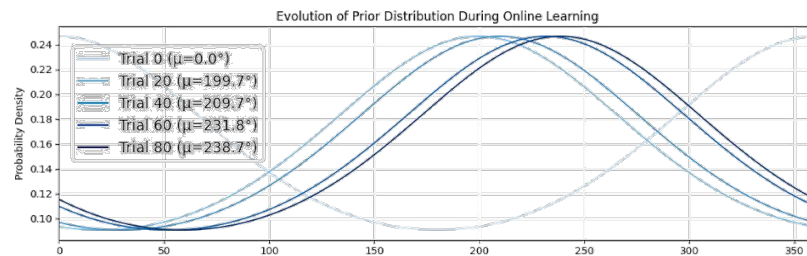
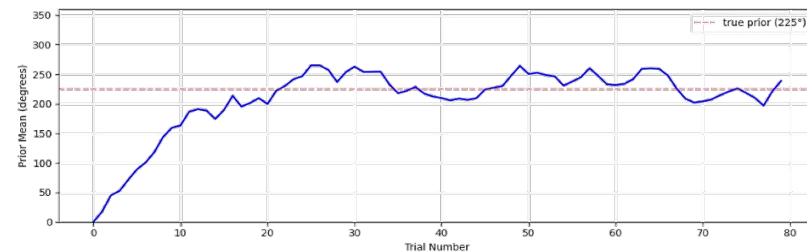
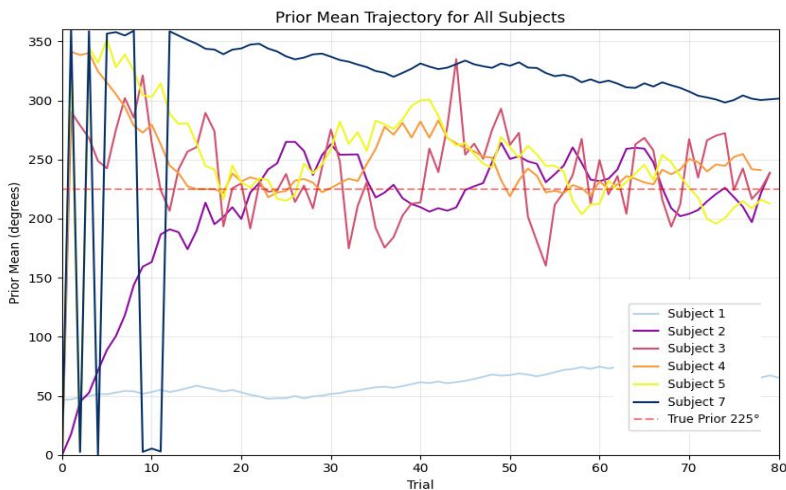
Online Bayesian Model

Prior: start from uniform Von-mises

$$\mu_{prior} += \alpha \times (\theta_{true} - \mu_{prior})$$

RMSE

subject	1	2	3	4	5	7
uniform	147.09	117.71	90.73	137.59	135.66	136.09
VM	77.48	46.94	59.72	69.78	61.25	69.40



Conclusion & Discussion

- **Summary**

- Implicit learning occurs despite high sensory uncertainty.
- Only a subset of participants showed clear learning trajectories.

- **Limitation**

- Behavioral data alone limits insight into the influencing factors.
- Model fitting was imperfect, with substantial noise remaining.

- **Future Ideas**

- Incorporating additional measures of engagement—such as eye tracking to clarify confounding factor.

